

Dancing Stick Figures: A Synthetic Video Dataset, Renderer, and Diagnostic Evaluation Suite

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Abstract

Training a video generation model is hard to learn by doing. Real video datasets are large, the process behind each frame is unknown, and one aggregate score rarely says why a generated video failed. **Dancing Stick Figures** is a synthetic video dataset with a deterministic renderer, built so that the whole experiment fits on one GPU. It contains 1,340 six-second motions rendered from three cameras: 4,020 videos and 482,400 labelled frames. Every frame stays linked to the joints, camera, body parameters, depth, surface normals, part labels, and source motion that produced it, so evaluation has an answer key. Rule-based measurements check whether coloured limbs stay connected and how position and pose change over time. Controlled corruptions test what each measurement can and cannot see; in a 120-frame study, playing videos backward leaves the time-symmetric motion signals unchanged and scores FVD 79.3 where real-real scores 80.5. The release includes a 0.85-GB 64² training tier, a 32² route, fixed evaluation manifests, checkpoints, and a Colab lesson that trains an image generator and then a video generator.

1 Introduction

Video generation is easy to encounter and hard to study. Systems such as Seedance and MiniMax H3 [11, 12] publish results and sometimes weights, but a learner still runs into three walls:

- **Data.** Without the training data, a reader cannot reconstruct the learning problem or trace a failure back to it. The exact data, filters, and recipe behind a strong model are rarely released together.
- **Experimentation.** Frontier training needs large corpora, heavy compute, and pretrained codecs and encoders. What remains affordable is prompting and fine-tuning. Retraining a controlled variant in an afternoon is not.
- **Evaluation.** “Better” can mean cleaner frames, an intact body, plausible motion, or prompt adherence, and these change independently. An aggregate score does not say which one changed or which component broke it.

The question this paper addresses is practical: can someone with a single GPU train video generation models from scratch on a small but meaningful dataset, and study their

behaviour through repeated, controlled experiments? That requires three things. The full image-to-video path must fit one consumer GPU or a Colab runtime. A run must produce inspectable multi-second samples in hours, not days. And evaluation must measure visible failures directly, with known blind spots.

Dancing Stick Figures is built to those constraints. It is a *dataset and renderer*: articulated motions whose rendered frames stay connected to their exact generating state. It is an *evaluation suite*: fixed manifests, rule-based structural measurements, and controlled corruptions that test the measurements themselves. And it ships reference models and a Colab workflow that trains an image generator, transfers its spatial weights to a video generator, trains it, samples multi-second videos, and diagnoses the failures.

The release contributes:

1. a label-rich video dataset with a 0.85-GB 64² tier, an optional 32² tier, the motion, camera, and body parameters behind every frame, and a deterministic renderer;
2. a seed-disjoint evaluation protocol, structural evaluator, and controlled corruption suite; and
3. image and video trainers, checkpoints, and an end-to-end Colab curriculum.

2 Related Work

Several lines of work provide synthetic video, human-motion supervision, or decomposed evaluation. Dancing Stick Figures combines data, renderer, reference models, and measurements in one experiment a reader can rerun and alter.

Small and controlled video. Moving MNIST [1] made temporal prediction cheap, but has no articulated structure or language conditioning. Kubric and MOVi [2] showed the value of programmable rendering with dense ground truth for scene understanding. Neither offers a compact route from text-conditioned generator training to failure diagnosis. We restrict the scene to one articulated figure so that the whole route stays small enough to rerun.

Synthetic people. SURREAL [3] and BEDLAM [4] render large, realistic human datasets and ask whether synthetic people can train perception systems. We ask a different question with a much smaller dataset: whether a

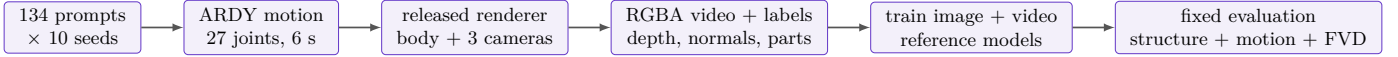


Figure 1: From source motion to diagnosis. Every rendered frame stays linked to the motion, camera, body, and buffers that generated it.

reader can train a video generator from scratch, observe its failures, and relate them back to known motion and rendering state.

Training data for video generation. HumanVid [5] and MiraData [6] improve the coverage and transparency of video training data at scale. They do not aim to make pretraining itself a short experiment. Our 0.85-GB tier makes the opposite trade: narrow visual scope for a pipeline that can be inspected end to end. Synthetic data has also been used to improve large generators directly [7, 8]; those works target a different compute regime, one where the reader inherits a large pretrained model rather than training the generator themselves.

Human motion as source data. AMASS [9] and HumanML3D [10] support motion modelling but specify no pixel renderer or visual evaluation. We use articulated motion as source material and simplify appearance so that generated pixels can be related to joints and limbs.

Evaluation of generated video. FVD [14] is a useful distributional distance, but it responds more to frame content than to temporal corruption [15]. T2V-CompBench and VBench-2.0 [23, 24] decompose quality into many axes; GeneVA [16] and Ref4D-VideoBench [17] localise artifacts with human or reference supervision. We work in one fully specified domain instead, where controlled mistakes show exactly which signals react and which do not. This complements semantic, perceptual, and human evaluation; it does not replace them.

3 Dataset and generation pipeline

We generated one motion per seed 0–9 for each of 143 prompts in six groups (dance, gesture, locomotion, transition, idle, sport) with NVIDIA ARDY [13]. Each motion is six seconds at 20 fps. A z-buffered capsule renderer with 4× supersampling emits three deterministic orthographic views. Nine prompts were removed by visual curation because their motions do not visibly perform the requested action in this domain: near-static failures (*sways to slow music*), motion the renderer cannot show (*shakes their head no*; the head is a filled circle), absent or unrecognisable actions (*yoga warrior pose*, *yoga tree pose*, *sit up*, *push up*, *standing long jump*), and interactions with objects that do not exist in the render (*climbs over a low wall*, *lies down on the floor*). The release contains 134 prompts; the exclusion list ships with per-prompt reasons. Prompts whose stillness is correct (the idle group) are kept even when the low-motion QA flag fires. Table 1 summarises

Table 1: Released data. A motion is rendered from three views.

Configuration	Rows	Resolution	Size
frames	482,400	128 ²	4.4 GB
mini	482,400	64 ²	0.79 GB
motion	1,340	27 joints	0.33 GB

Table 2: Seed-disjoint splits. Each split contains all 134 prompts; source motions are disjoint by seed.

Split	Prompts	Motions	Videos	Frames
Train	134	1,072	3,216	385,920
Validation	134	134	402	48,240
Test	134	134	402	48,240
All	134	1,340	4,020	482,400

the data. Seeds 0–7 train, seed 8 validates, seed 9 tests: every prompt appears in all three splits, and no source motion crosses a split boundary (Table 2).

The generator samples limb length, scale, stroke, and camera yaw and pitch deterministically from `clip_id`. It stores 3D and projected joints, visibility, camera and body parameters, RGBA, 16-bit depth, camera-space normals, 27-part segmentation, and the raw motion (Figure 2). The domain uses one visual style, one figure per frame, orthographic cameras, and at most 128² resolution. The builder records two motion-quality flags (**frozen**, **levitation**); flagged samples stay in the release so filtering choices are explicit.

The configurations are compute tiers of one dataset. 128² keeps the most detail; the 0.85-GB 64² tier is the primary training tier; 32² supports smoke tests and rapid ablations. Reported structural results use 64², where the thin coloured limbs remain measurable.

The motion release supports rebuilding the full visual dataset without ARDY: a script re-derives body and camera settings from each clip identifier and reruns the public renderer. A verifier checks metadata exactly and compares rendered colour, segmentation, visibility, depth, and normals within declared tolerances.

4 Tasks and failure diagnostics

Video task. Each clip has 120 frames at 20 fps, and dataset-level evaluation uses the complete clip. The reference models train on a declared window view: 40 consecutive frames at a random offset within the first 64 frames of each clip. The 40-frame length matches ARDY’s two-second generation window and keeps the training tensor small; the first-64-frame restriction concentrates training where the prompted action occurs, because ARDY does not pace an action to the requested duration. The window rows in Table 3 compare generated windows with real

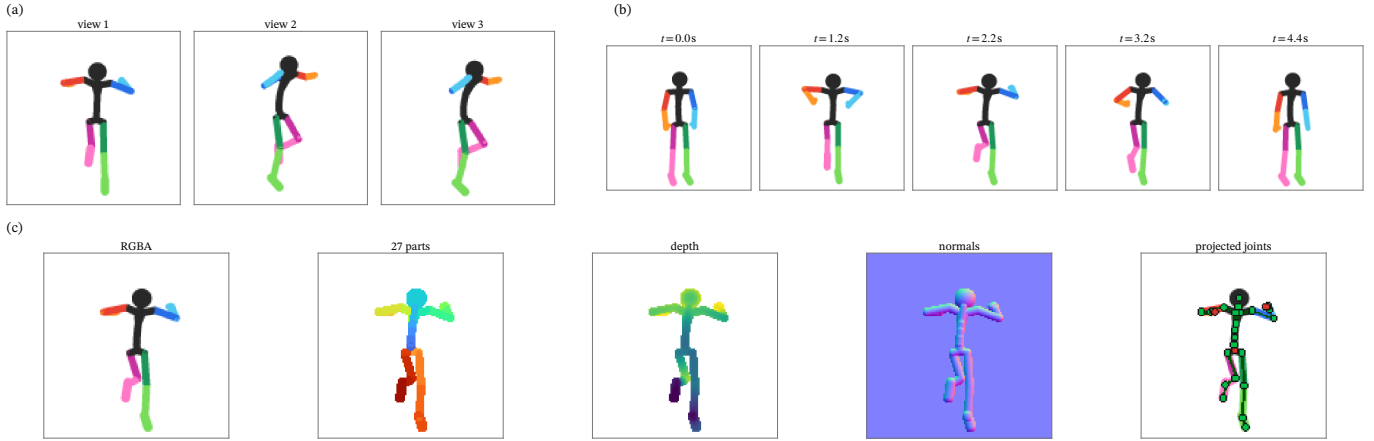


Figure 2: One released motion viewed through the dataset. (b) Five frames at equal increments of pose-space motion, so an idle tail cannot dominate the strip. (a) The middle pose from the clip’s three cameras. (c) The same frame with part segmentation, depth, camera-space normals, and projected joints. Each record also stores its prompt, source-motion identifier, seed, camera, and body parameters.

windows drawn under the same manifest rule.

Structural answer key. Each arm and leg uses a fixed pair of colours. The evaluator converts a frame into eight limb-colour masks and reports:

- **Topology violation rate (TVR):** limb colours missing or split into multiple components;
- **Limb-identity error (LIE):** expected colour adjacencies that are absent;
- **Colour-purity error (CPE):** foreground pixels outside the rendering palette;
- **Foreground fraction:** pixels occupied by the figure; and
- **Clean rate:** frames with both TVR and LIE at zero.

For video, it adds colour-mass drift, centroid speed and acceleration, motion fraction, angular speed and jerk, and height variation. These are image-space diagnostics, not biomechanics. The motion quantities are two-sided reference signals: zero can mean a stopped video, and large values can mean jitter. Agreement with the real reference, not a small value, is the target.

Occlusion produces non-zero topology scores even on ground-truth renders, so ground-truth scores are a *real reference*, not an error floor. A deterministic manifest selects one camera per source motion and keeps the two 128-video reference halves motion-disjoint. For FVD [14] we composite RGBA onto white, resize to 224^2 , and extract I3D features; values are comparable only within this fixed implementation and sample count.

Controlled validation. Five deterministic structural corruptions were applied to 500 held-out rerenders. A partial left/right limb swap raises LIE by .203; an extra arm raises TVR by .085. A whole-limb swap preserves the colour graph and is not detected; bone stretching and hand deletion are also missed, because the metrics measure topology, not proportions or end-effectors. Temporal corruptions (freeze, shuffle, loop, reverse) are evaluated in

§5.

5 Diagnostic validation and reference results

Metric stress test. Figure 3 reports the 120-frame study. The two real sets have centroid speeds .313 and .299 pixels per frame. Freezing zeroes all motion signals. Shuffling raises centroid acceleration from .348 to 2.858 and angular jerk from .060 to .384; looping raises them to .507 and .084. Reversal changes neither the frame set nor any reported motion signal, and FVD scores it 79.3 against the real reference’s 80.5 (freeze 465.5, shuffle 419.2, loop 328.5). Across 30 paired 64-video trials, reversal changes FVD by -1.65 ± 2.69 . Reversal is a shared blind spot: the hand-designed signals are time-symmetric by construction, and FVD does not see it either. Neither measurement certifies directionally correct motion.

Released reference models. The reference experiments form a ladder that changes one variable per rung. The shared backbone is a 39.9M-parameter factorised video DiT: 4×4 frame patches, alternating spatial and temporal attention blocks, flow-matching velocity prediction (a linear interpolant; often called rectified flow, though we apply no reflow step) [19, 20], and a frozen T5-small encoder attached through cross-attention. With $T=1$ the network is an image generator; with $T=64$ it generates the full training window jointly.

The first rung is initialisation: the 64-frame model trains from random weights and, separately, warm-started from the image checkpoint’s spatial and text weights. Image-first initialisation is established practice [21, 22]; the two rows are reference points, and the random row is the matched comparator for the latent rows. The second rung varies the mechanism that carries motion across patch boundaries: the factorised baseline, a zero-initialised local $3 \times 3 \times 3$ con-

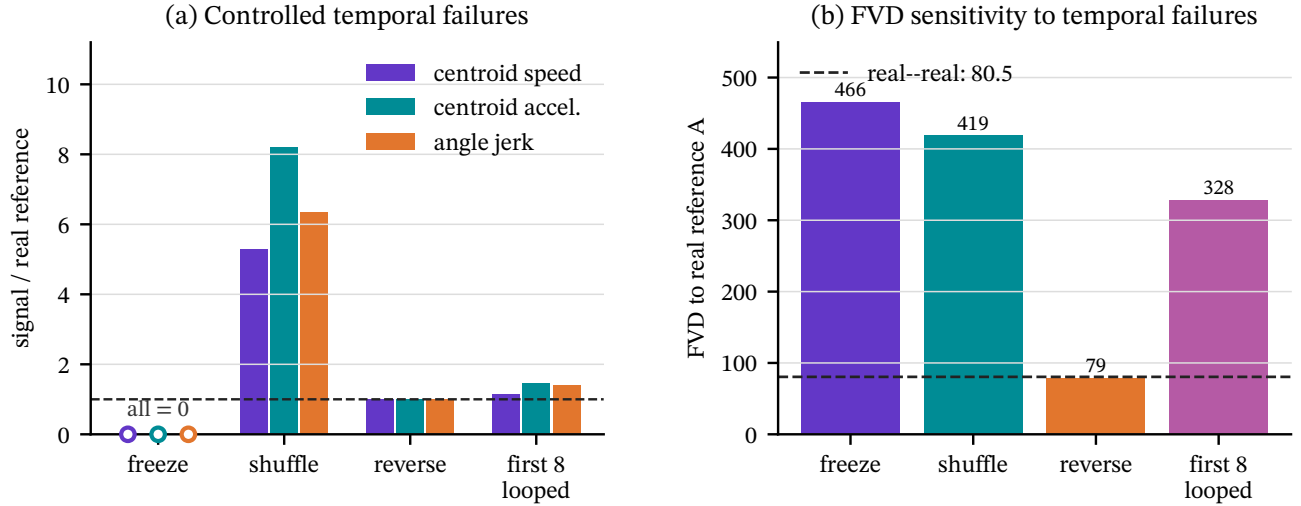


Figure 3: Controlled study on complete six-second clips ($n = 128$ per set). *Freeze* repeats frame 1, *shuffle* permutes all 120 frames, *reverse* plays them backward, *first 8 looped* repeats frames 1–8. (a) Freezing zeroes the motion signals; shuffling and looping create excess acceleration and angular jerk; reversal changes neither. (b) Against the 80.5 real–real reference, FVD responds to freezing, shuffling, and looping, but scores reversed video at 79.3.

volitional mixer added to every block, and joint attention over all spatio-temporal tokens. These are measured at pixel scale with the same warm start; joint attention costs $4.7\times$ the factorised step in wall-clock. The latent block repeats the two attention extremes where 1,024 tokens make them nearly cost-matched (0.17 vs. 0.19 s/step). The third rung varies representation: matched DiTs trained in the latent space of a separately trained frozen video autoencoder, decoded before evaluation, with a codec-floor row (real windows encoded and decoded by the same codec) to separate codec cost from model skill.

The codec predates the v0.2 curation and split, and a post-hoc audit found two defects: it reconstructs training foregrounds with 63% lower masked error than held-out foregrounds, and neutralising its background latent cells degrades foreground reconstruction $33\times$ (a re-trained reference codec: $3\text{--}5\times$). The codec memorised training clips and stores foreground information in background latents. The latent rows therefore illustrate the representation axis under a weak codec; they do not bound what latent training can reach. Re-trained codecs that pass this audit (held-out gap $\approx 10\%$) are planned for a revision.

Table 3 collects the runs. Each block names its protocol, because single-frame and 64-frame scores are different measurements. Earlier v0.1 baselines remain released but were trained under a superseded split and are not comparable rows. Figure 4 shows the warm-started model.

Reading the ladder. Initialisation dominates at this budget. From random weights the 10k-step video model violates topology in 94% of frames (TVR .942, FVD 1641); warm-started, the same model reaches TVR .169 against .116 real. The seed repeat (.183, 570.7) shows these contrasts are large against training-seed variance. The two measurement families also disagree in opposite directions.

The warm-started pixel model has near-real structure but too much motion (centroid speed .850 vs. .373 real), which FVD punishes. The latent rows are the opposite: FVD near 500, TVR above .8, broken figures rendered smoothly by the codec. Ranking by TVR alone and by FVD alone gives opposite orderings. Changing the prompt under fixed noise has a measurable effect, and the paired wave examples in Figure 4 move the requested arm. This is bounded evidence of conditioning, not a calibrated prompt-adherence score.

Why pixel space. At $32^2\text{--}64^2$, pixel-space training is practical and keeps thin-limb errors in the representation the evaluator scores. A separately trained video VAE adds a training stage and can confound generator errors with codec artifacts, as the audited codec above shows. The core Colab therefore follows the pixel-space route.

6 Accessibility and reproducibility

The repository releases the generator, cache builder, trainers, sampling code, evaluator, corruptions, FVD evaluation, checkpoints, and comparison script. The notebook inspects the data, builds a cache, trains the image generator, transfers its spatial weights, trains the video generator, samples a multi-second GIF, and compares scores against real references. 32^2 is a smoke test; 64^2 is the reference exercise. Each stage leaves an inspectable artifact before the next begins.

The motion release supports a full rebuild. The rebuild script recomputes body and camera settings from each `clip_id` and reruns the public renderer; ARDY is not required. The verifier was run over all 514,800 128^2 frames of the pre-curation generation: every motion and metadata label matches; colour and segmentation pixels agree on

Table 3: Reference ladder on the released seed-disjoint data. Every model trains on the same 64^2 cache and first-64-frame view with the same flow-matching trainer, optimiser, and step budget; one variable changes per rung. Rows are comparable only within a protocol block. “Image” initialisation copies compatible spatial and text weights from the single-frame checkpoint.

Model	Attention	Init.	TVR↓	LIE↓	CPE↓	Speed	Motion frac.	Jerk	FVD↓
<i>Single frames, prompt-conditioned, $n = 512$, 50 sampling steps</i>									
Real validation frames	—	—	.115	.093	.039	—	—	—	—
Image DiT, 30k	factorised	random	.128	.050	.030	—	—	—	—
<i>64-frame windows at 20 fps, prompt-conditioned, $n = 128$ held-out source motions</i>									
Real reference windows	—	—	.116	.093	.037	.373	.501	.073	115.2 / ref.
Codec reconstruction floor	—	—	.137	.104	.037	.380	.503	.104	124.4
Video DiT, 10k	factorised	random	.942	.035	.051	.408	.559	.326	1641.3
Video DiT, 10k	factorised	image	.169	.045	.034	.850	.719	.324	545.0
Video DiT, 10k	factorised + local mixer	image	—	—	—	—	—	—	—
Video DiT, 10k	joint spatio-temporal	image	—	—	—	—	—	—	—
Latent video DiT, 10k	factorised	random	.816	.398	.084	.571	.647	.276	503.3
Latent video DiT, 10k	joint spatio-temporal	random	.821	.408	.087	.564	.635	.273	527.2

TVR is topology violation rate, LIE limb inconsistency error, CPE colour-purity error. Speed, motion fraction, and jerk are two-sided signals: agreement with the real row is the target. The codec-floor row passes the real windows through the frozen autoencoder used by the latent rows; latent scores should be read against it. Real training-split windows score FVD 125.7 against the same reference, which is also what a training-set replay would score, so models near this band need a memorisation check. A training-seed repeat of the warm-started video row scores TVR .183 / FVD 570.7 against .169 / 545.0.

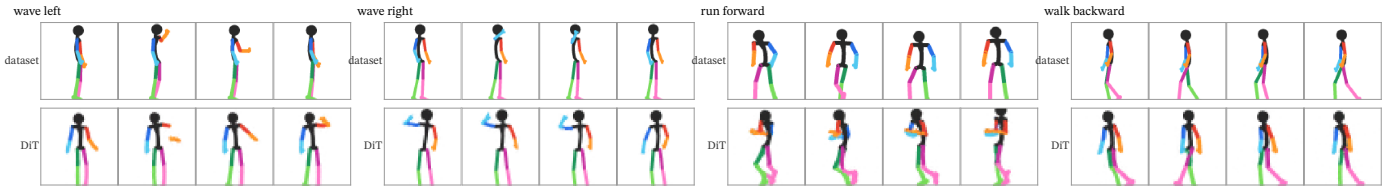


Figure 4: Released clips and 10k-step DiT samples. Each block interleaves four time points from a released clip and a generation (50 Euler steps, guidance 3). The paired wave prompts check laterality; running and backward walking show locomotion under independently seeded noise.

99.88% and 99.92% of comparisons; joint visibility, the least stable field, agrees on 99.50%; depth and normals exceed 99.99%. An independent 64^2 rebuild passes with similar agreement. The same script can re-render with seeded overrides to body scale, stroke, or camera, and the manifest records the overrides; an instructor keeps the private seed for a course-specific rendering.

Resource cost decides who can run the experiment, so we report it with the results. On the ladder’s GPU (NVIDIA RTX PRO 6000 Blackwell, 96 GB): the 30k-step image stage runs at 0.10 s/step with a 10.5 GB peak; each 10k-step 64-frame video run peaks below 10 GB with gradient checkpointing and fits a 12 GB consumer card. Measured step times are 3.4 s for the factorised run (two runs sharing the GPU) and 1.07 s for the local-mixer run (exclusive). Disabling checkpointing raises the local-mixer peak to 55.1 GB and lowers its step to 0.82 s; that trade only pays on large workstation cards. Step times transfer across GPUs only approximately. The earlier UNet lesson completed on a hosted Tesla T4; hardware numbers are reported for the hardware they were recorded on.

Our code is MIT and the dataset is CC0-1.0. Motions were generated with ARDY’s 20-fps Core model (checkpoint revision not retained). ARDY’s code is Apache-2.0; its checkpoints are distributed under the [NVIDIA Open Model Agreement](#), which permits derivative works and claims no ownership of outputs.

7 Limitations and next evaluations

The domain is narrow by design: one raster style, one figure, orthographic cameras, no real-video proxy. The colour-based evaluator measures visible topology, not bone length, joint angles, left/right semantics, or biomechanics. The motions inherit ARDY’s capabilities and biases; the nine-prompt curation removed what inspection found, but we have not human-validated every remaining motion, and ARDY does not pace actions to the requested duration. The seed-disjoint test measures in-domain generation under unseen motion realisations, not zero-shot prompts or human motion at large. The colour metrics do not establish prompt adherence. Three sampling seeds quantify generation noise for one checkpoint, not training variation or scaling.

The most direct extension is a learned rig estimator with calibrated per-joint confidence: map a generated video back to a rig, re-render it, and compare. That would expose rig-space failures (bone length, joint angles, foot contact) and rendered-space disagreement, but its scores would need validation on controlled malformations before use as evidence.

8 Conclusion

Dancing Stick Figures makes a complete video-generation experiment small enough to run and concrete enough to inspect. The released workflow trains an image genera-

tor, warm-starts a video generator from it, samples multi-second videos, and connects visible failures to measurements of body-part connectivity and motion. Controlled mistakes show why the measurements must be read together: the structural signals miss reversed time, and FVD does not name the property that failed. It is a small, complete first experiment in video generation.

Use of AI systems. The source motions were generated with ARDY’s 20-fps Core model (§5); rendering, curation, and verification are deterministic code. An AI coding assistant (Claude) was used for experiment orchestration, evaluation tooling, and manuscript editing under author direction. All numerical results come from the released code and were reviewed by the author.

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